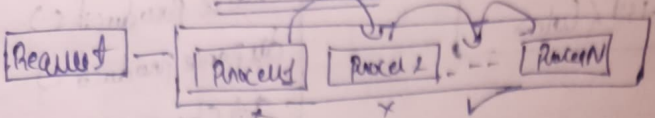
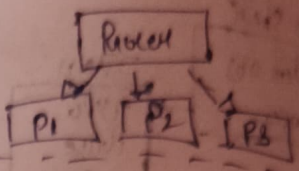
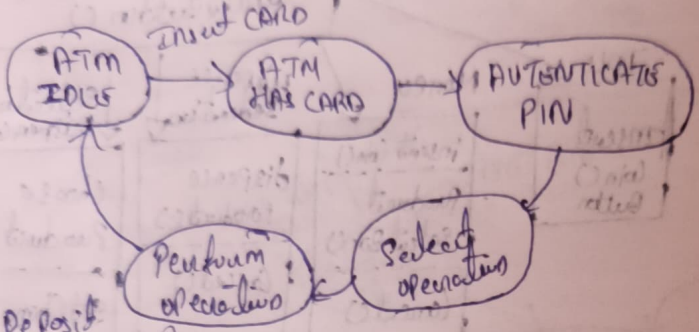
(9) DESIGN ATM = STATE & CHAIN OF RESPONSIBILITY.

→ COR: Behavioural pattern, allows multiple handlers to process a request sequentially.

If one handler cannot process, it passes to next handler







WHERE STATE PATTERN USED:

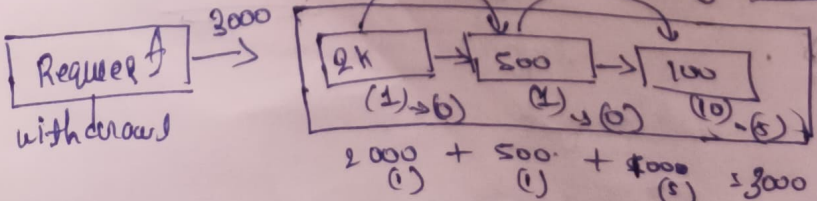
Operations:
1. Insert Card
2. Authenticate PIN
3. Select option
4. Perform option

STATE
1. IDLE → Insert CARD
2. HASCARD → Authenticate PIN
3. Select option → Choose option
4. Perform → DO Internal operation

→ IDLE state can only insert card → HASCard can only validate PIN

WHERE CHAIN OF RESPONSIBILITY USED

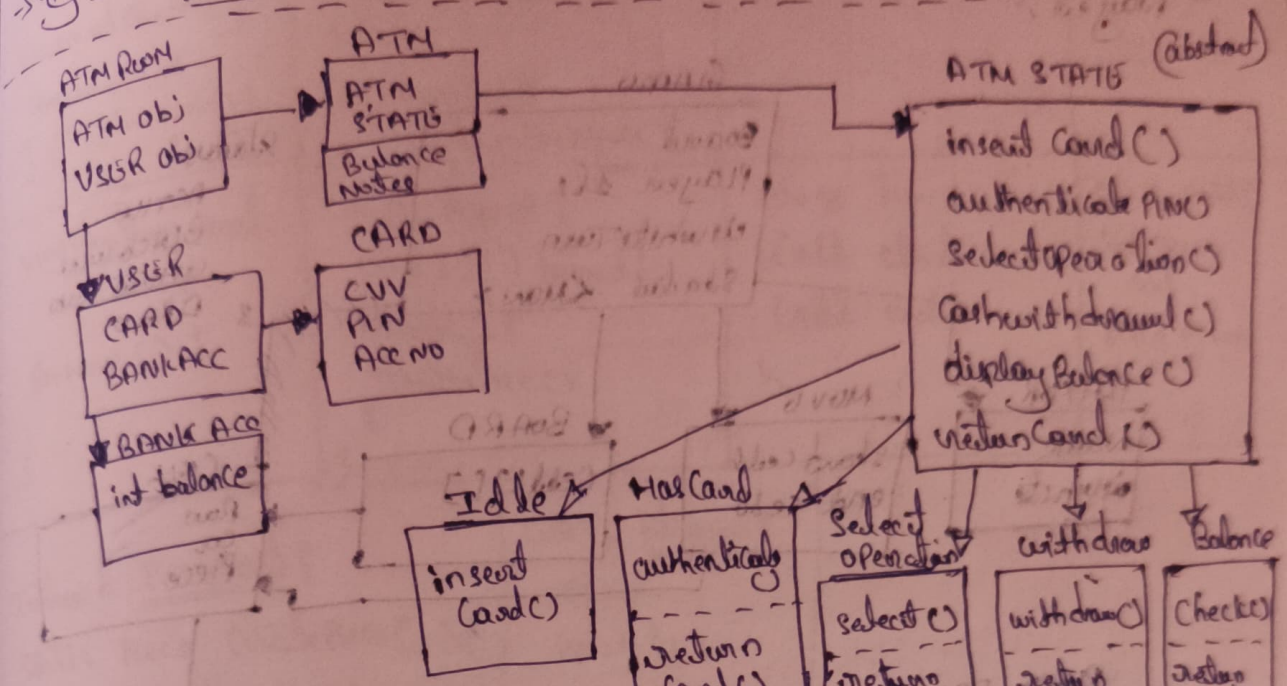
→ WITHDRAWL OPERATION: we know we have 3 type of note 2K, 500, 100. we get in descending order.



Request will go to each handler & if it cannot fulfil go to next

* Note: Dispense it with Interfaces, how do we need to handle WITHDRAWL.

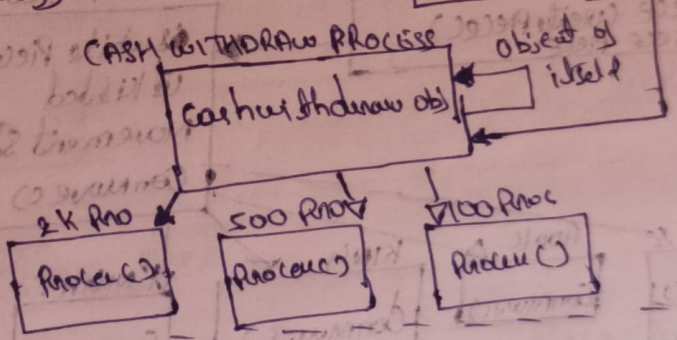
→ If simple one, don't use COR else use COR.



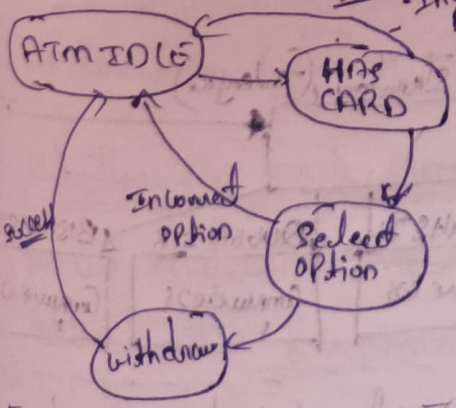
(new 200()(new 500()(new 100())

2000 → 500 → 100.

Process accordingly.



FLOW



① Create ATM obj, USER obj, CARD obj, BANKACC obj.

② Create ATM ROOM, assign ATM & USER objects.

③ STATE = IDLE → insertCARD().

④ STATE = ~~Authenticate~~ HASCARD → authenticatePIN().

⑤ STATE = SELECTION → select option. exit withdrawal

⑥ withdraw state → withdrawal() → Idle state

Call COR from withdrawal Process